

Report

	soapdenovo2	spades_illumina	spades_hifi	spades_clr	spades_ont	canu_purge	wtdbg2	nextdenovo	flye	mecat2	necat	hifiasm	quickmerge
# contigs (>= 0 bp)	1673	2212	786	697	773	21	237	19	33	54	23	48	17
# contigs (>= 1000 bp)	194	170	92	76	85	21	237	19	33	54	23	48	17
# contigs (>= 5000 bp)	126	126	70	66	65	21	217	19	32	43	23	48	17
# contigs (>= 10000 bp)	108	114	60	62	58	21	152	19	31	36	23	48	17
# contigs (>= 25000 bp)	91	96	54	54	51	20	85	19	26	24	21	29	17
# contigs (>= 50000 bp)	72	77	48	47	46	20	64	19	25	21	19	18	17
Total length (>= 0 bp)	11808089	11776373	11865834	11894785	11863211	11930927	11208242	12117195	12644004	12719391	12089579	12944107	11649126
Total length (>= 1000 bp)	11544506	11493952	11692830	11748361	11694449	11930927	11208242	12117195	12644004	12719391	12089579	12944107	11649126
Total length (>= 5000 bp)	11402099	11393278	11649452	11727966	11647990	11930927	11119815	12117195	12640402	12688999	12089579	12944107	11649126
Total length (>= 10000 bp)	11275750	11303464	11576266	11695626	11596340	11930927	10661928	12117195	12633028	12637696	12089579	12944107	11649126
Total length (>= 25000 bp)	10966740	10963472	11465063	11588188	11463627	11917900	9612923	12117195	12534019	12426077	12046009	12575589	11649126
Total length (>= 50000 bp)	10255303	10292428	11245634	11317421	11276045	11917900	8860527	12117195	12500212	12328204	11972414	12240638	11649126
# contigs	260	217	135	101	121	21	237	19	33	54	23	48	17
Largest contig	451640	559760	656089	742337	964465	1098218	386018	1077585	1506186	1109999	1372365	1538968	1530938
Total length	11590536	11527224	11721007	11764908	11717889	11930927	11208242	12117195	12644004	12719391	12089579	12944107	11649126
Reference length	12157105	12157105	12157105	12157105	12157105	12157105	12157105	12157105	12157105	12157105	12157105	12157105	12157105
GC (%)	38.08	38.09	38.09	38.09	38.09	38.13	37.68	38.54	38.97	38.02	38.64	38.36	38.55
Reference GC (%)	38.15	38.15	38.15	38.15	38.15	38.15	38.15	38.15	38.15	38.15	38.15	38.15	38.15
N50	155146	132528	309451	346331	418442	746639	148997	803374	708187	820950	840809	785250	919267
NG50	150198	131788	309451	315700	312850	746639	138801	803374	736036	820950	840809	807453	848201
N90	43989	45985	97202	100738	99751	354807	16594	434960	222105	269434	268150	271987	434960
NG90	28128	26112	82619	81652	83573	319503	6683	434960	435282	438998	268150	440952	337371
auN	167890.3	178367.2	335588.5	351202.2	382421.6	748395.5	146688.3	787967.8	765014.0	732996.1	809472.0	812991.1	931675.2
auNG	160066.0	169125.7	323550.3	339872.1	368605.3	734471.9	135239.2	785381.1	795653.3	766898.4	804975.8	865620.9	892745.6
L50	26	27	13	12	11	7	27	7	7	7	6	7	5
LG50	27	29	13	13	12	7	30	7	6	7	6	6	6
L90	76	79	39	38	37	15	108	14	17	17	15	16	12
LG90	91	96	44	42	42	16	188	14	15	15	15	14	13
# misassemblies	1	4	9	14	3	36	44	22	37	45	26	80	26
# misassembled contigs	1	4	7	7	3	12	33	10	17	19	9	38	9
Misassembled contigs length	697	859719	2040213	728856	1162761	8014234	3358044	5993917	8713016	5085302	5971222	9282963	6602108
# local misassemblies	0	0	0	1	0	7	3	0	12	2	5	7	3
# scaffold gap ext. mis.	0	0	0	1	0	0	0	0	0	0	0	0	0
# scaffold gap loc. mis.	138	0	1	3	7	0	0	0	0	0	0	0	0
# unaligned mis. contigs	0	0	0	0	0	0	9	0	3	0	0	0	0
# unaligned contigs	3 + 1 part	4 + 0 part	4 + 0 part	3 + 0 part	3 + 0 part	1 + 9 part	74 + 107 part	0 + 0 part	4 + 27 part	1 + 4 part	0 + 3 part	1 + 14 part	0 + 3 part
Unaligned length	5935	6590	6610	5719	5719	24330	1639920	0	512580	29568	11502	38227	11813
Genome fraction (%)	94.949	94.758	96.274	96.399	96.210	96.903	76.281	99.806	99.248	99.383	99.273	99.390	95.671
Duplication ratio	1.001	1.000	1.001	1.002	1.000	1.010	1.031	0.998	1.006	1.048	0.997	1.065	0.998
# N's per 100 kbp	536.53	19.55	4.60	177.10	50.19	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
# mismatches per 100 kbp	1.77	4.66	4.88	8.32	6.55	37.79	327.20	31.81	458.03	44.33	62.71	38.27	49.78
# indels per 100 kbp	7.81	1.82	1.69	1.96	1.94	5.03	704.28	853.70	1747.60	1177.59	1066.46	8.65	951.24
Largest alignment	451537	468361	656089	742337	743798	1051571	384014	1077579	1059252	1108369	1077346	1051571	1077346
Total aligned length	11556075	11518701	11712199	11737092	11701512	11899035	9560948	12105713	12135893	12661002	12038101	12863964	11612730
NA50	154993	126937	261288	315700	312850	520240	135939	726728	524471	547114	559209	511385	728598
NGA50	150198	125255	260518	312841	309451	520240	107082	726728	533282	547114	559209	538500	658121
NA90	43065	42957	87257	100738	96946	203865	-	226852	129672	186609	256832	82453	226627
NGA90	28128	26112	72204	81652	83573	196978	-	226852	203745	230177	227521	216385	187608
auNA	167435.2	169646.7	305137.0	345591.0	352732.8	534407.5	127980.8	663468.7	552466.4	590201.5	611883.7	510204.7	636411.4
auNGA	159632.1	160857.0	294191.2	334442.0	339989.2	524465.0	117991.9	661290.7	574593.0	617499.3	608485.1	543233.3	609819.2
LA50	26	28	14	12	12	9	29	8	9	8	8	10	7
LGA50	27	30	15	13	13	9	33	8	8	8	8	9	8
LA90	77	82	44	39	39	22	-	18	25	24	19	27	19
LGA90	91	98	49	43	43	23	-	18	22	21	20	22	21

All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).

Misassemblies report

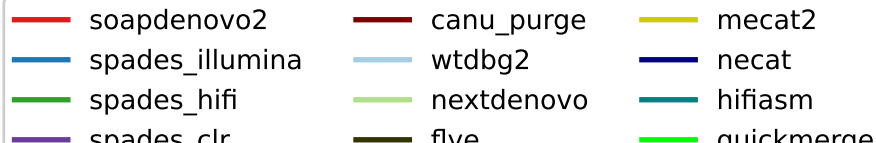
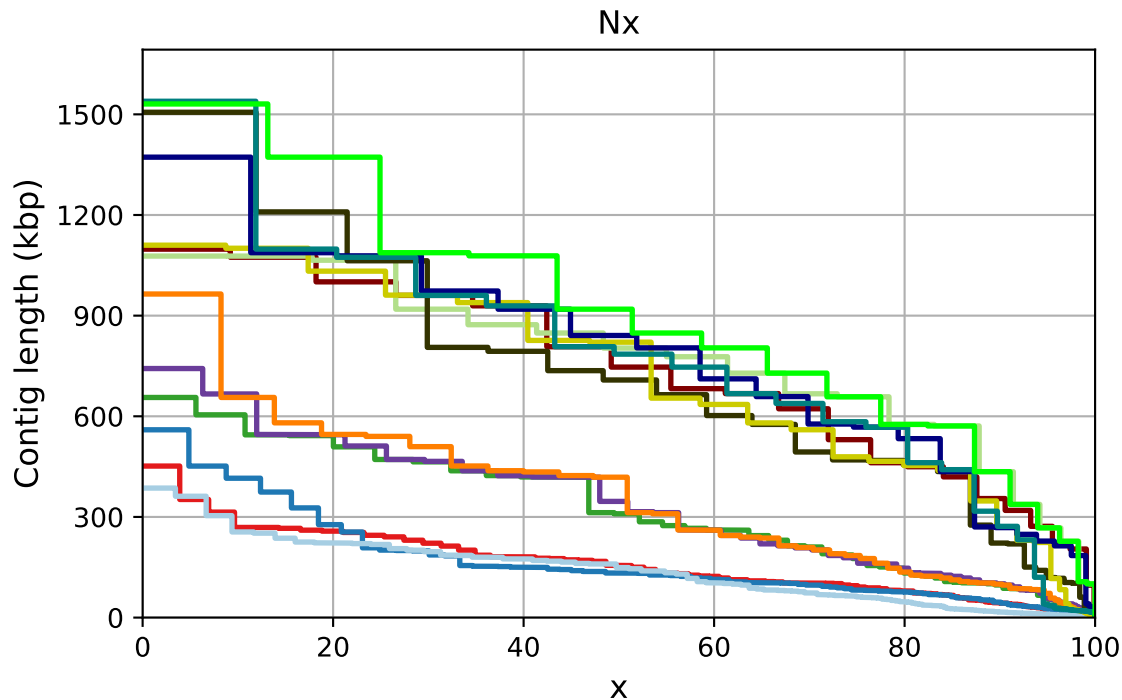
	soapdenovo2	spades_illumina	spades_hifi	spades_clr	spades_ont	canu_purge	wtdbg2	nextdenovo	flye	mecat2	necat	hifiasm	quickmerge
# misassemblies	1	4	9	14	3	36	44	22	37	45	26	80	26
# contig misassemblies	0	4	9	6	3	36	44	22	37	45	26	80	26
# c. relocations	0	2	5	3	1	18	37	9	18	30	14	61	12
# c. translocations	0	2	4	3	2	18	6	13	19	14	12	19	14
# c. inversions	0	0	0	0	0	0	1	0	0	1	0	0	0
# scaffold misassemblies	1	0	0	8	0	0	0	0	0	0	0	0	0
# s. relocations	0	0	0	4	0	0	0	0	0	0	0	0	0
# s. translocations	1	0	0	0	0	0	0	0	0	0	0	0	0
# s. inversions	0	0	0	4	0	0	0	0	0	0	0	0	0
# misassembled contigs	1	4	7	7	3	12	33	10	17	19	9	38	9
Misassembled contigs length	697	859719	2040213	728856	1162761	8014234	3358044	5993917	8713016	5085302	5971222	9282963	6602108
# local misassemblies	0	0	0	1	0	7	3	0	12	2	5	7	3
# scaffold gap ext. mis.	0	0	0	1	0	0	0	0	0	0	0	0	0
# scaffold gap loc. mis.	138	0	1	3	7	0	0	0	0	0	0	0	0
# unaligned mis. contigs	0	0	0	0	0	0	9	0	3	0	0	0	0
# mismatches	205	537	571	977	766	4497	31283	3851	55586	5613	7549	4923	5781
# indels	902	210	198	230	227	598	67336	103346	212087	149095	128381	1113	110465
# indels (<= 5 bp)	211	155	167	190	189	519	67181	101040	207643	146036	125212	990	107892
# indels (> 5 bp)	691	55	31	40	38	79	155	2306	4444	3059	3169	123	2573
Indels length	23505	6595	4041	5415	4173	4751	84124	170990	351532	228501	221287	6326	186136

All statistics are based on contigs of size ≥ 500 bp, unless otherwise noted (e.g., "# contigs (≥ 0 bp)" and "Total length (≥ 0 bp)" include all contigs).

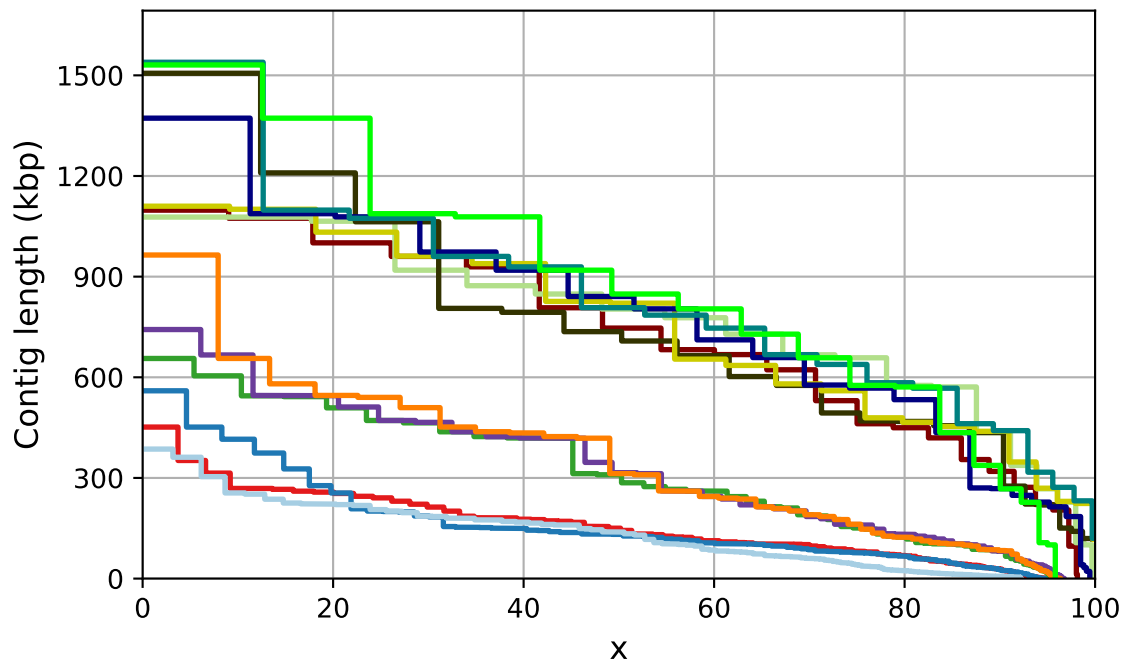
Unaligned report

	soapdenovo2	spades_illumina	spades_hifi	spades_clr	spades_ont	canu_purge	wtdbg2	nextdenovo	flye	mecat2	necat	hifiasm	quickmerge
# fully unaligned contigs	3	4	4	3	3	1	74	0	4	1	0	1	0
Fully unaligned length	5292	6590	6610	5719	5719	13027	734736	0	53668	24378	0	21604	0
# partially unaligned contigs	1	0	0	0	0	9	107	0	27	4	3	14	3
Partially unaligned length	643	0	0	0	0	11303	905184	0	458912	5190	11502	16623	11813
# N's	62187	2253	539	20836	5881	0	0	0	0	0	0	0	0

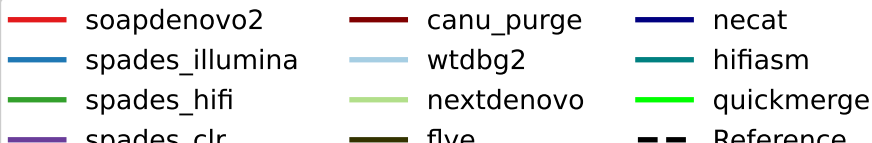
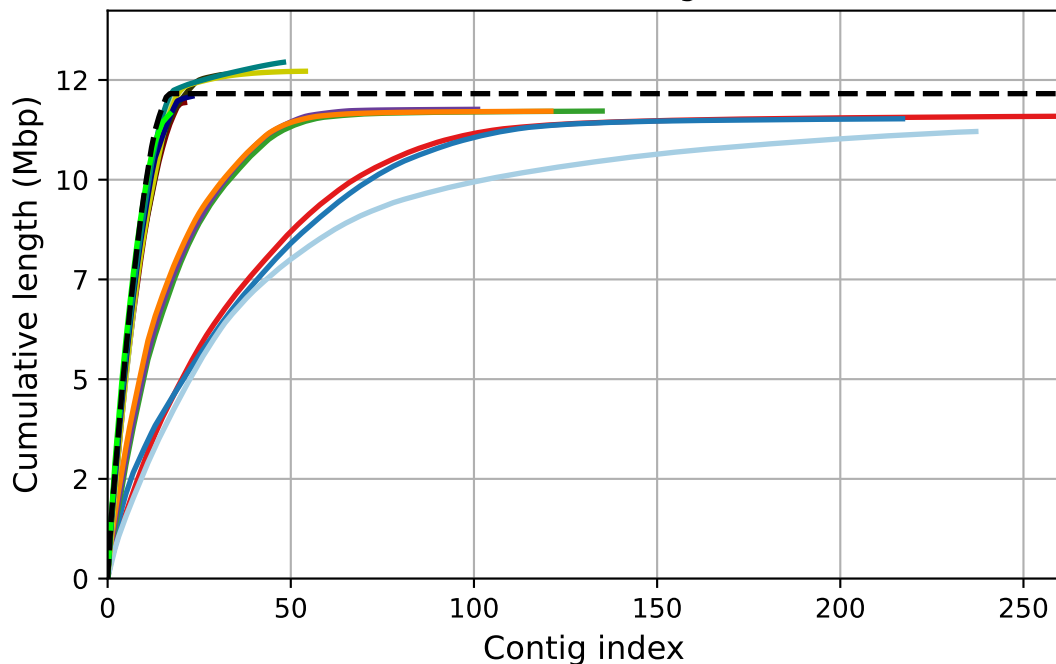
All statistics are based on contigs of size ≥ 500 bp, unless otherwise noted (e.g., "# contigs (≥ 0 bp)" and "Total length (≥ 0 bp)" include all contigs).



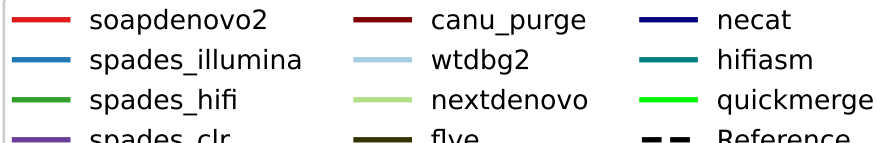
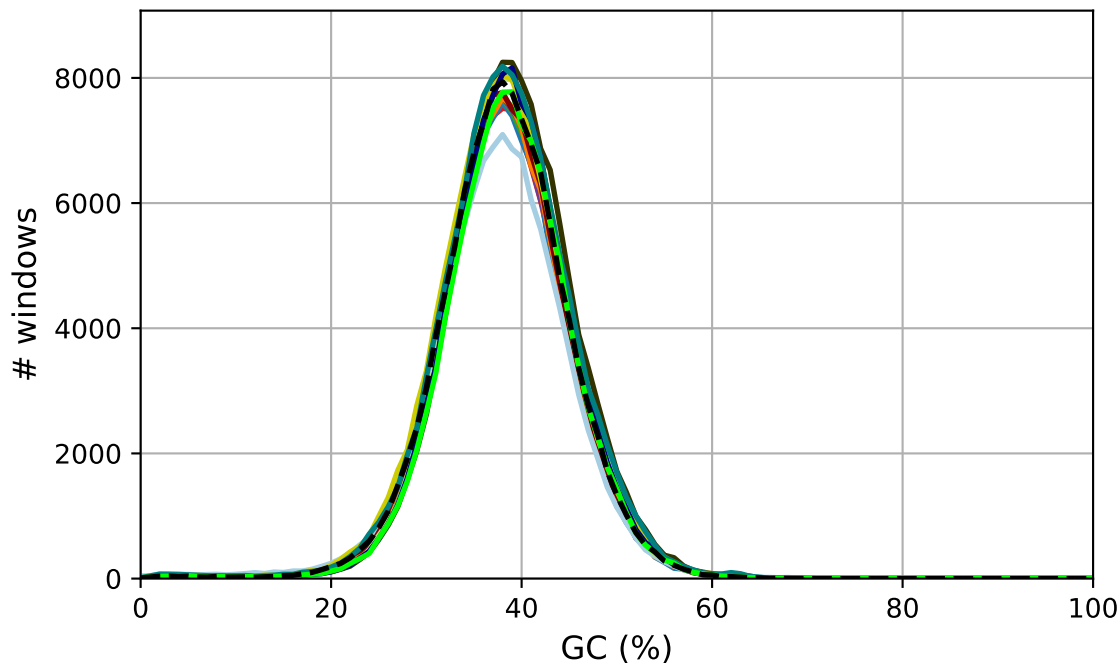
NGx



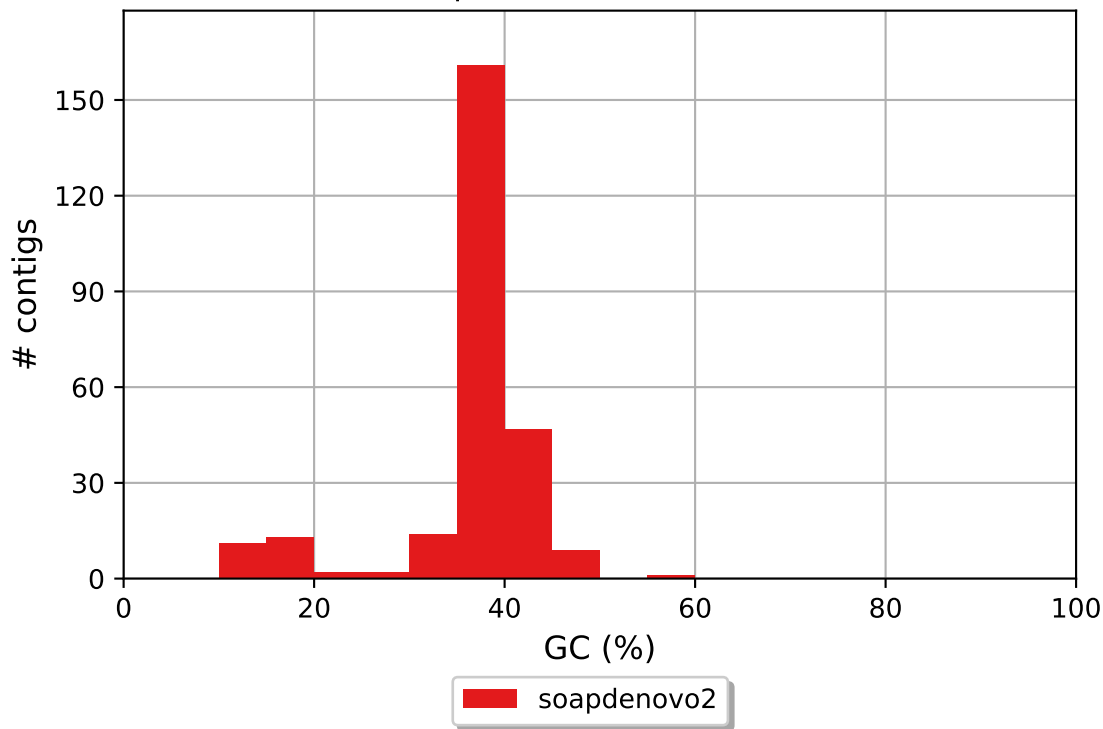
Cumulative length



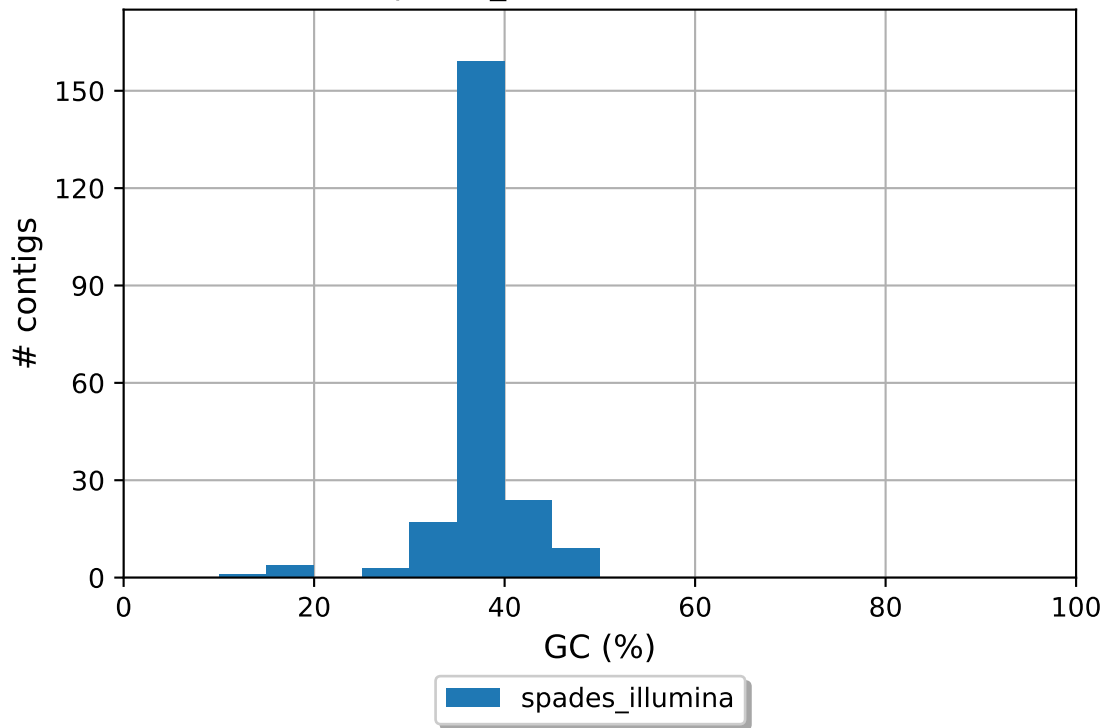
GC content



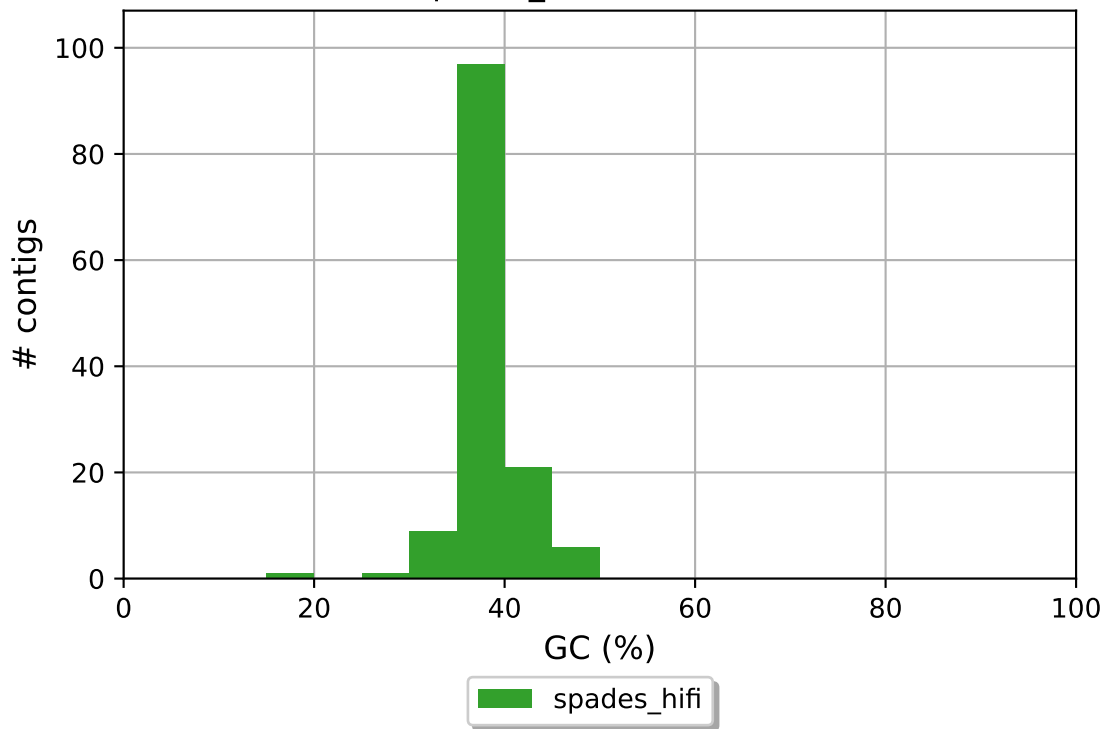
soapdenovo2 GC content



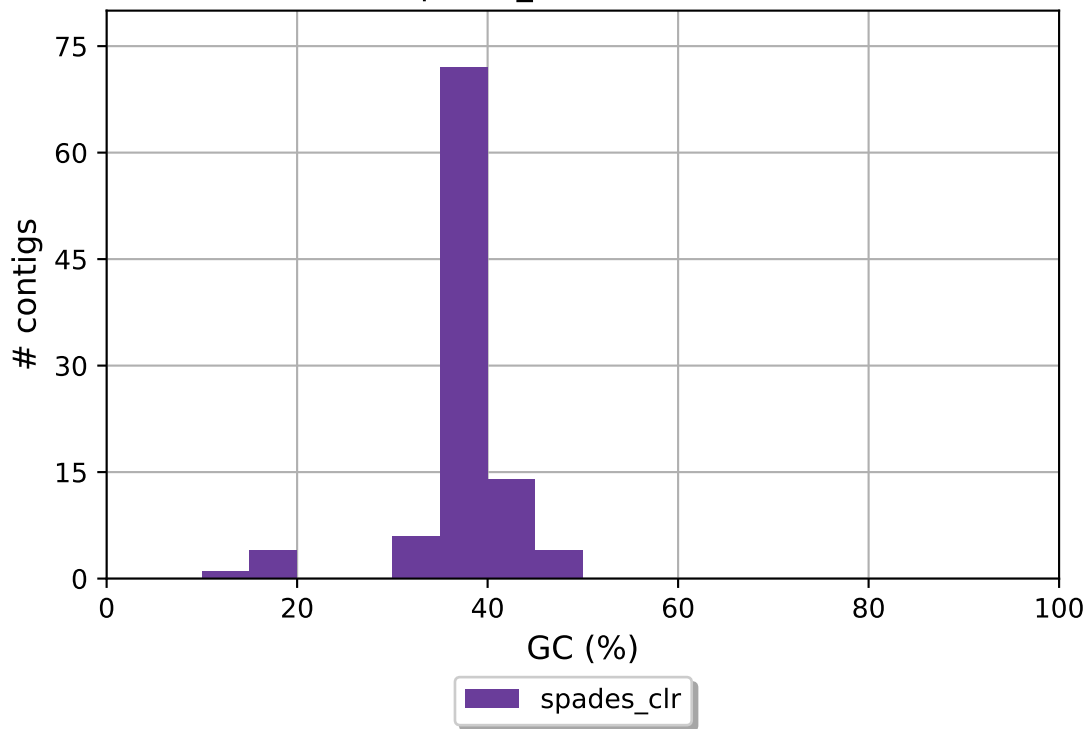
spades_illumina GC content



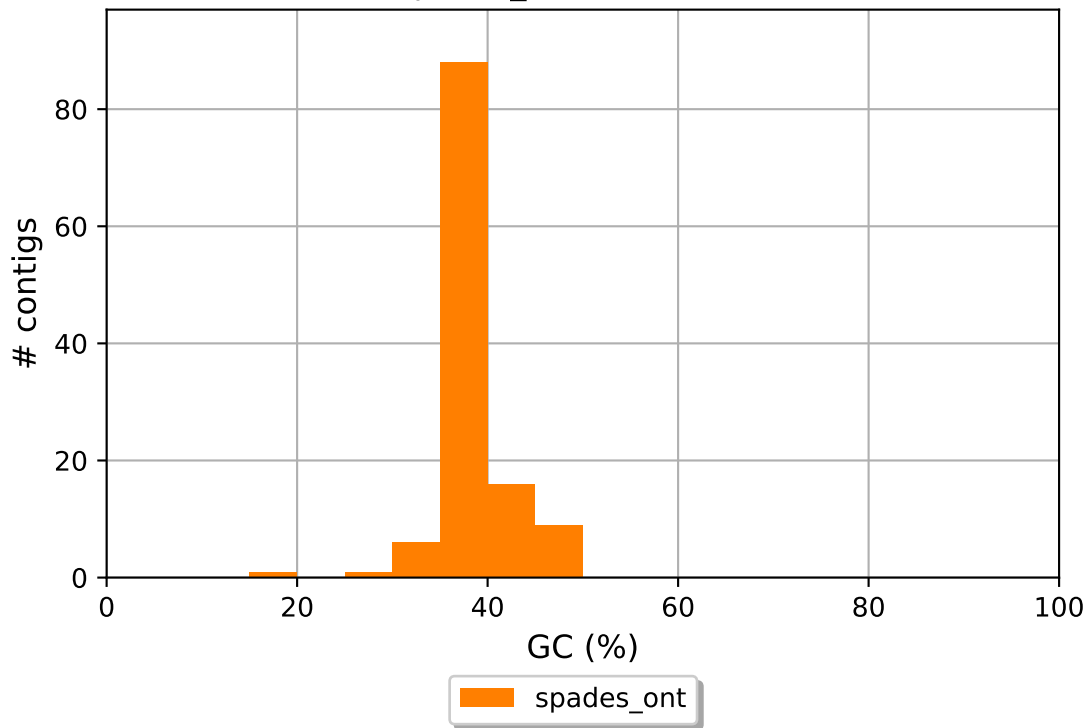
spades_hifi GC content



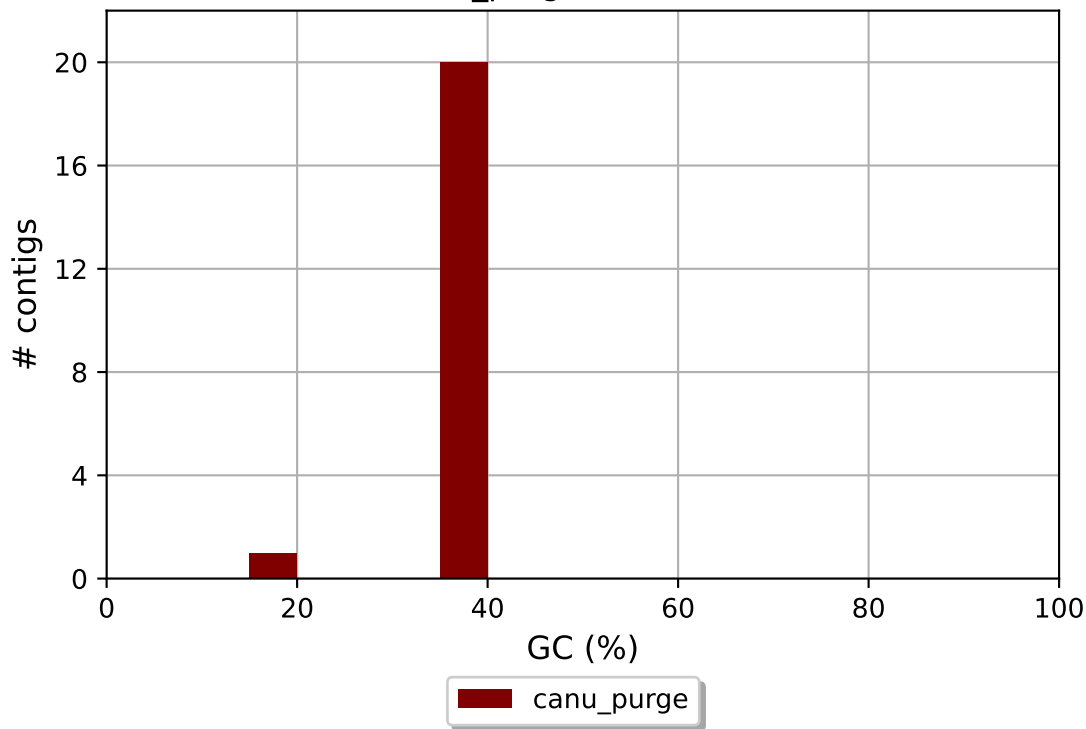
spades_clr GC content



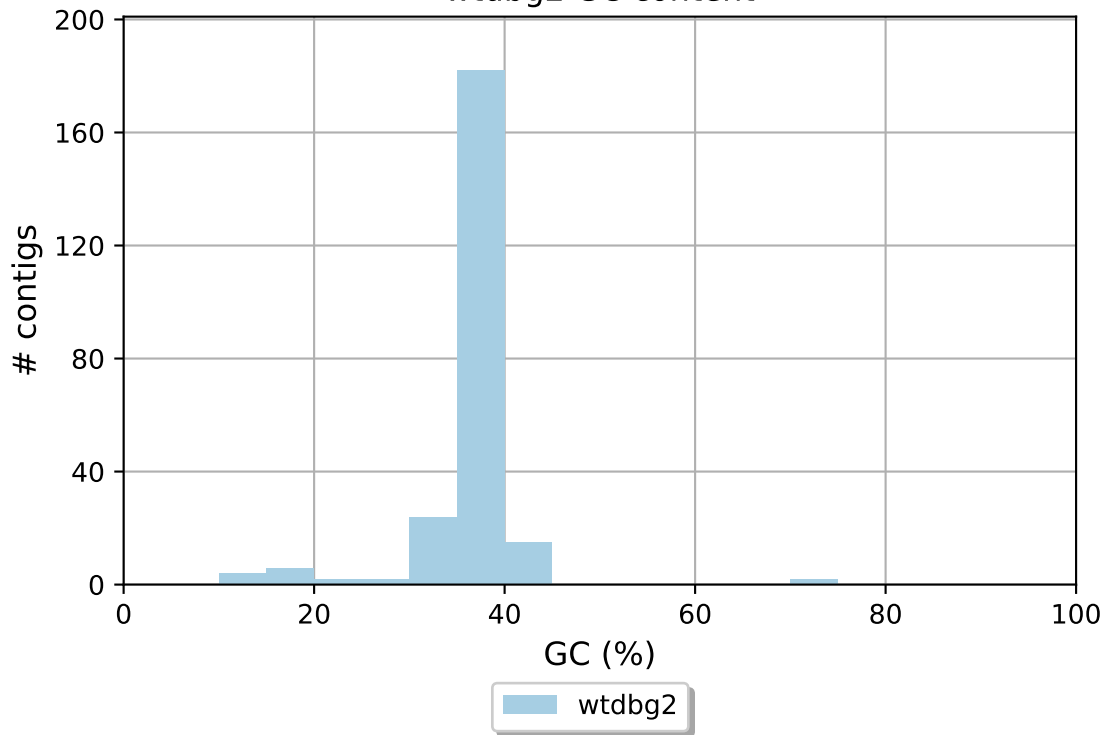
spades_ont GC content



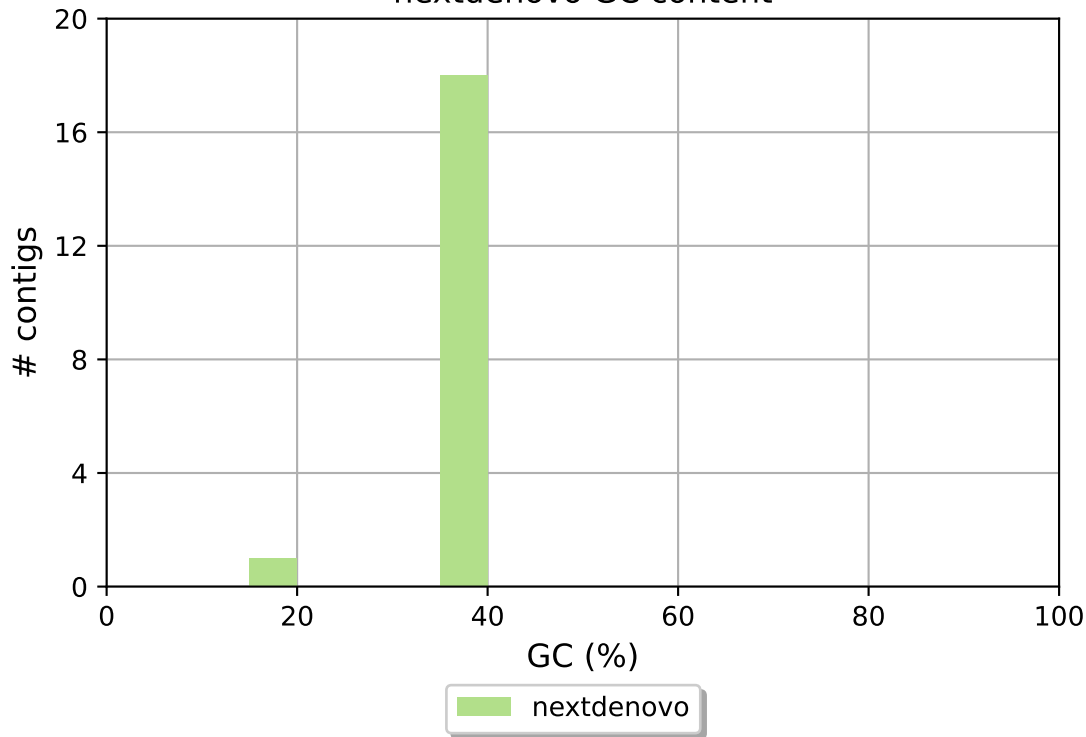
canu_purge GC content



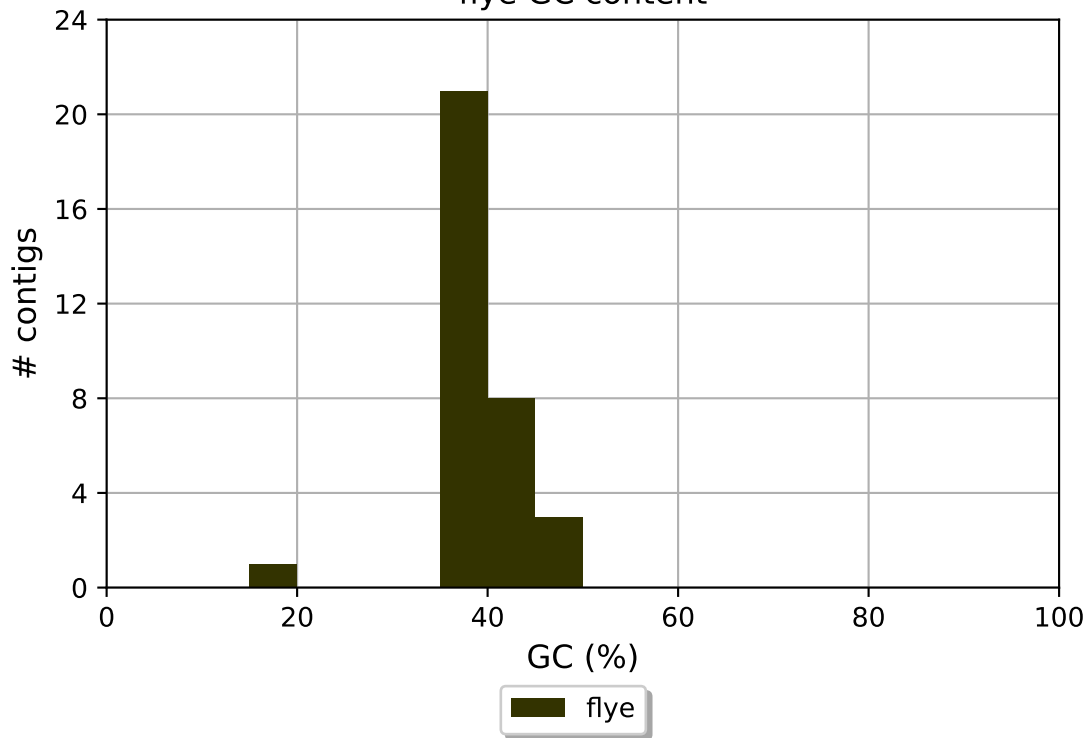
wtdbg2 GC content



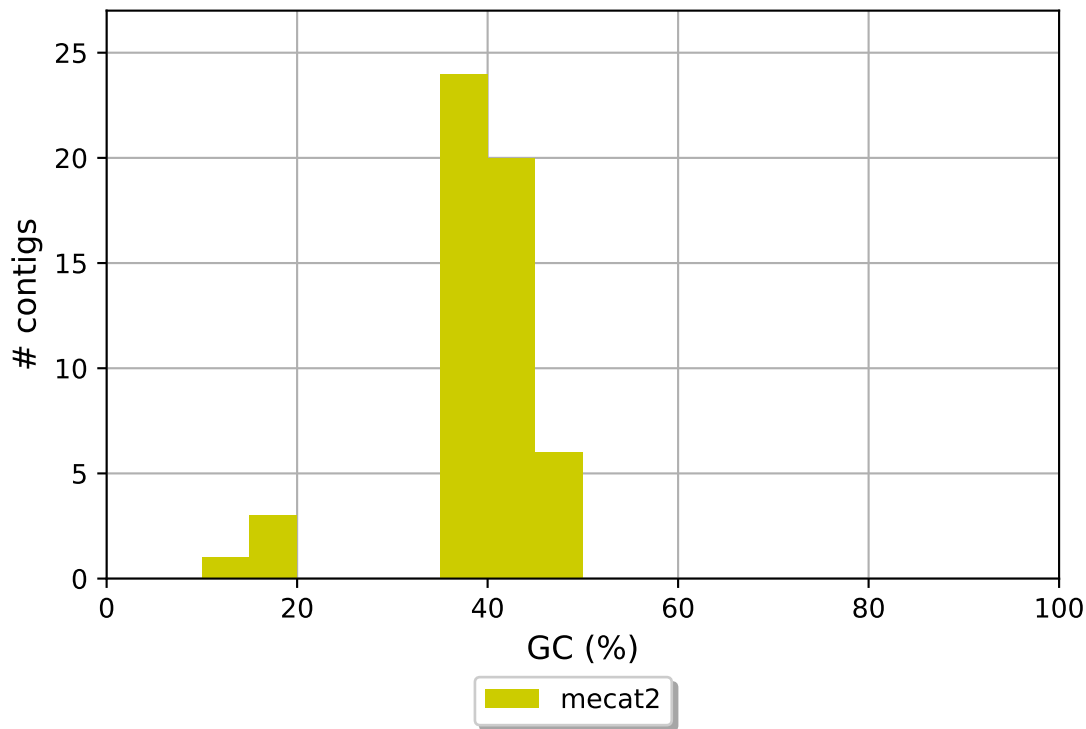
nextdenovo GC content



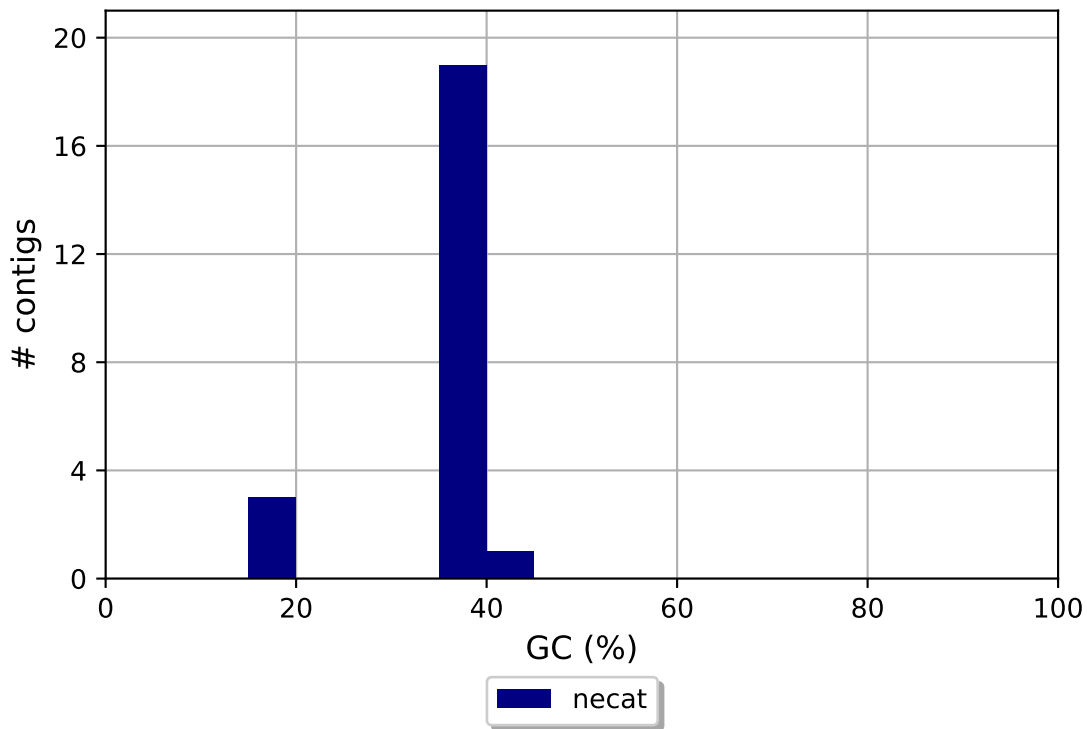
flye GC content



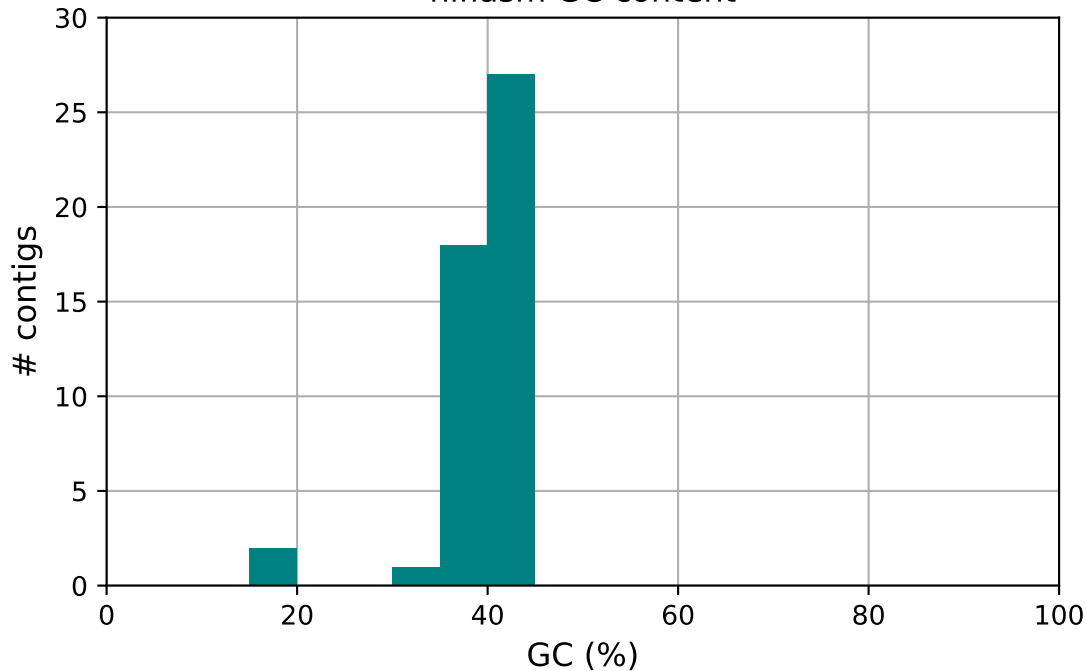
mecat2 GC content



necat GC content

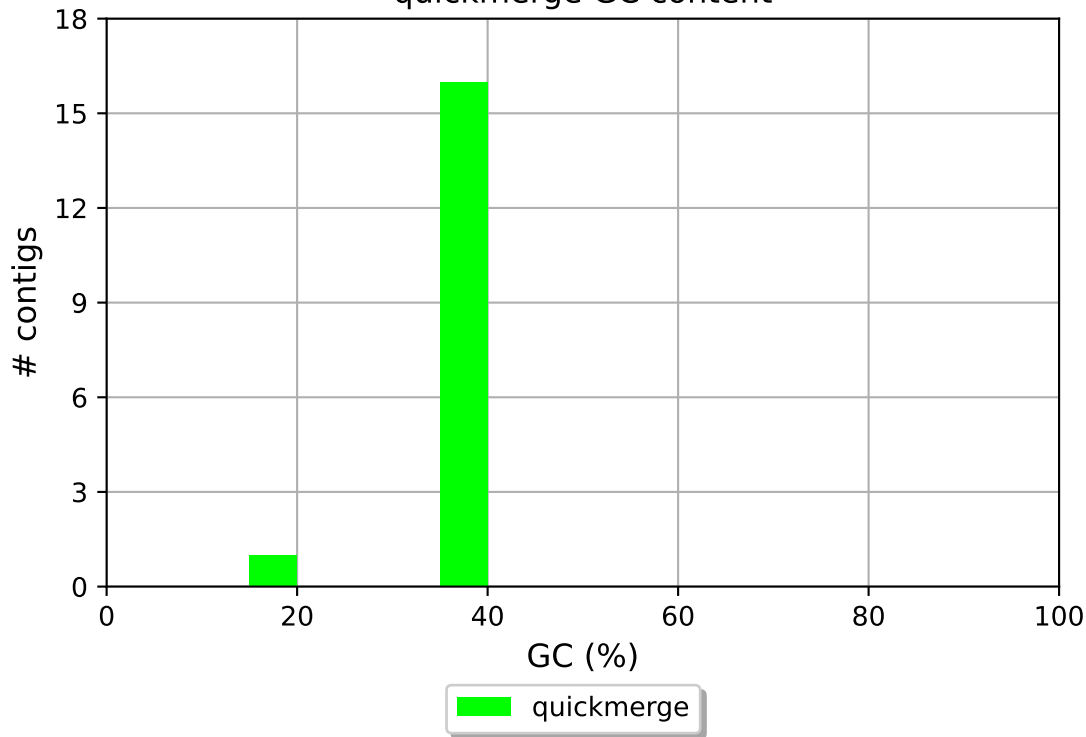


hifiasm GC content

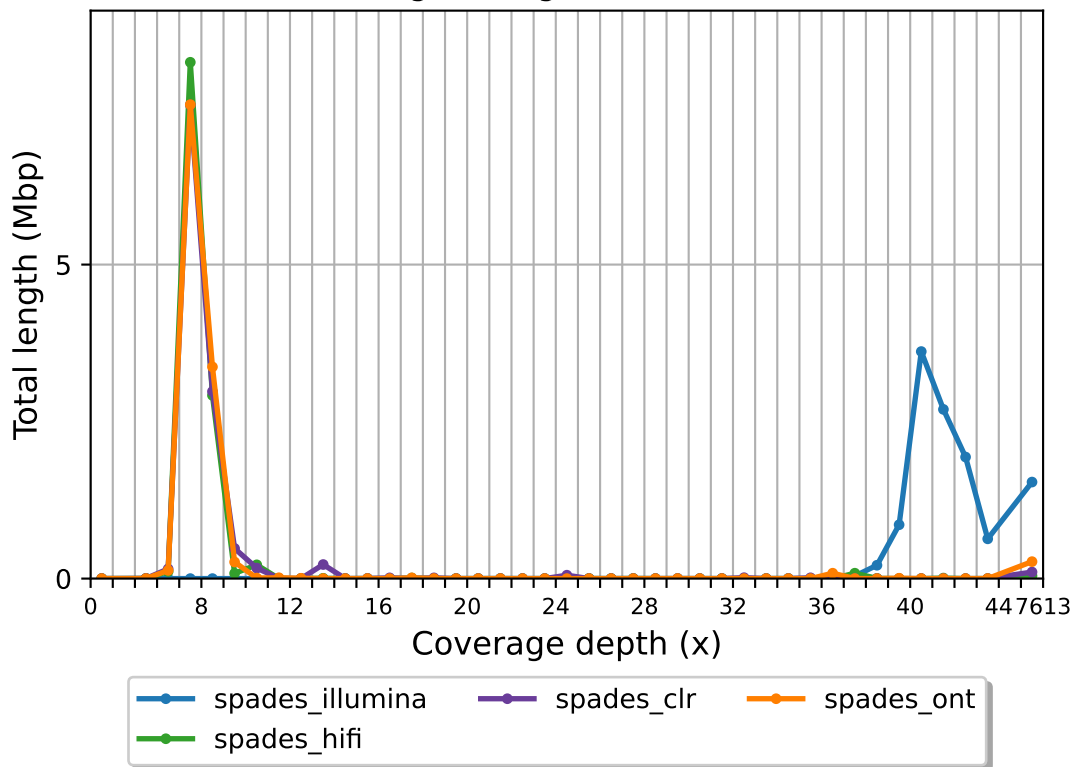


hifiasm

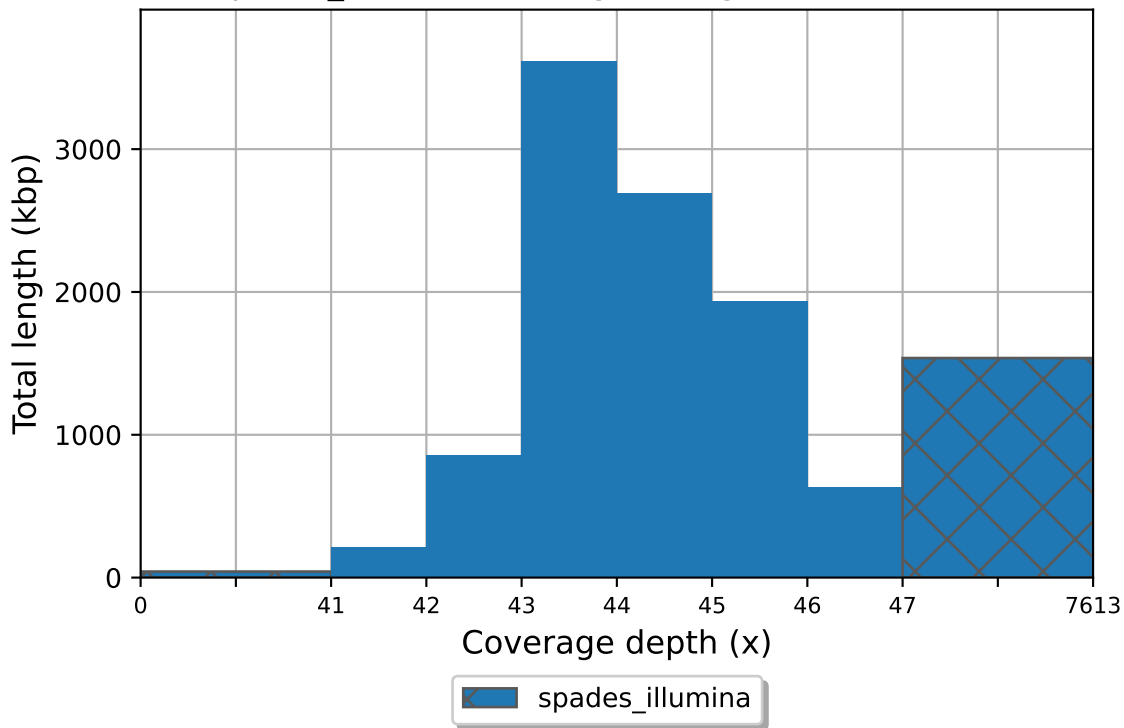
quickmerge GC content



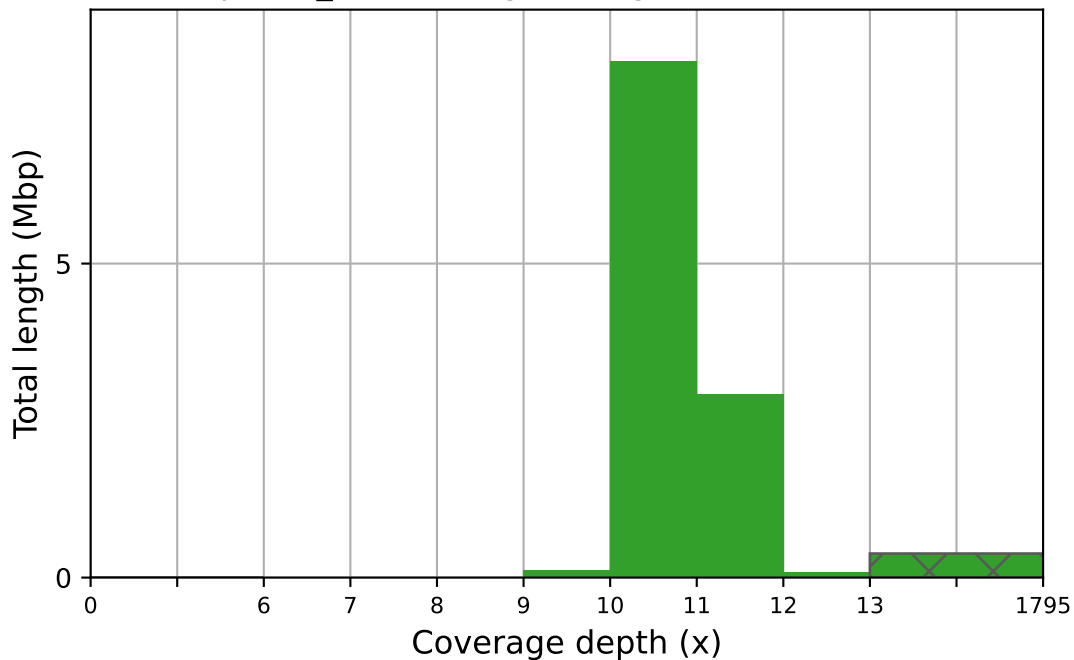
Coverage histogram (bin size: 1x)



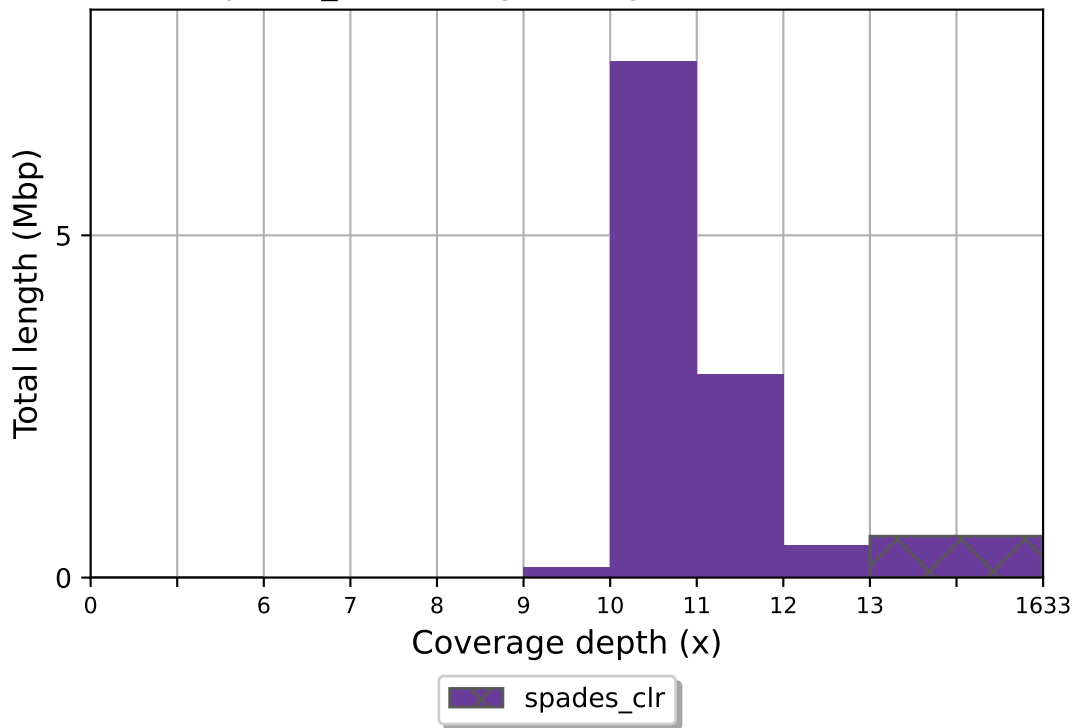
spades_illumina coverage histogram (bin size: 1x)



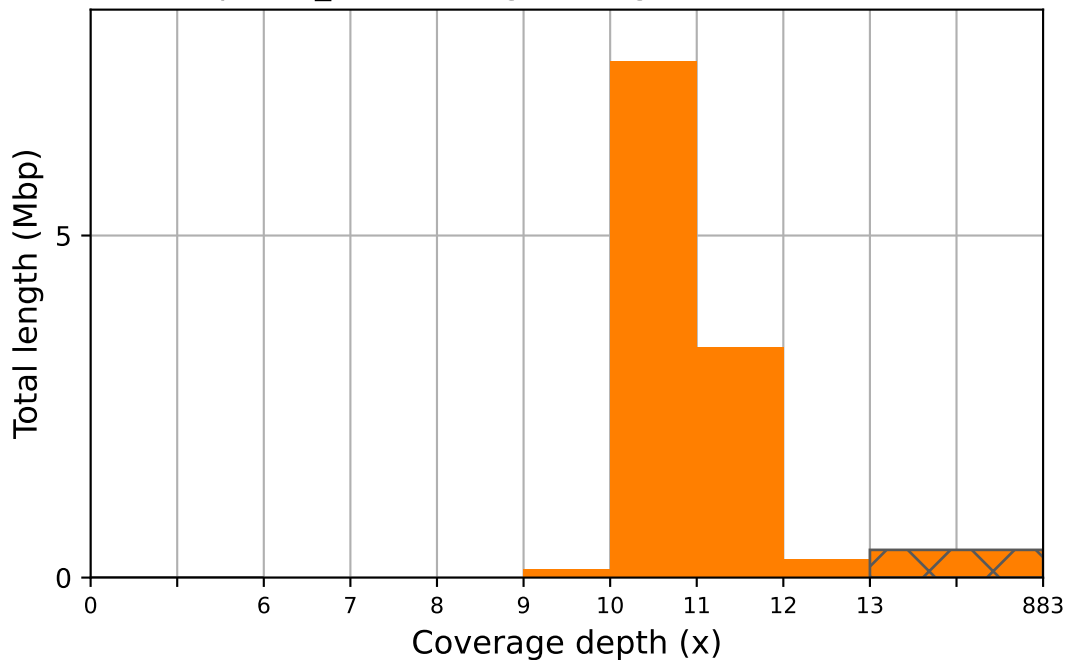
spades_hifi coverage histogram (bin size: 1x)



spades_clr coverage histogram (bin size: 1x)

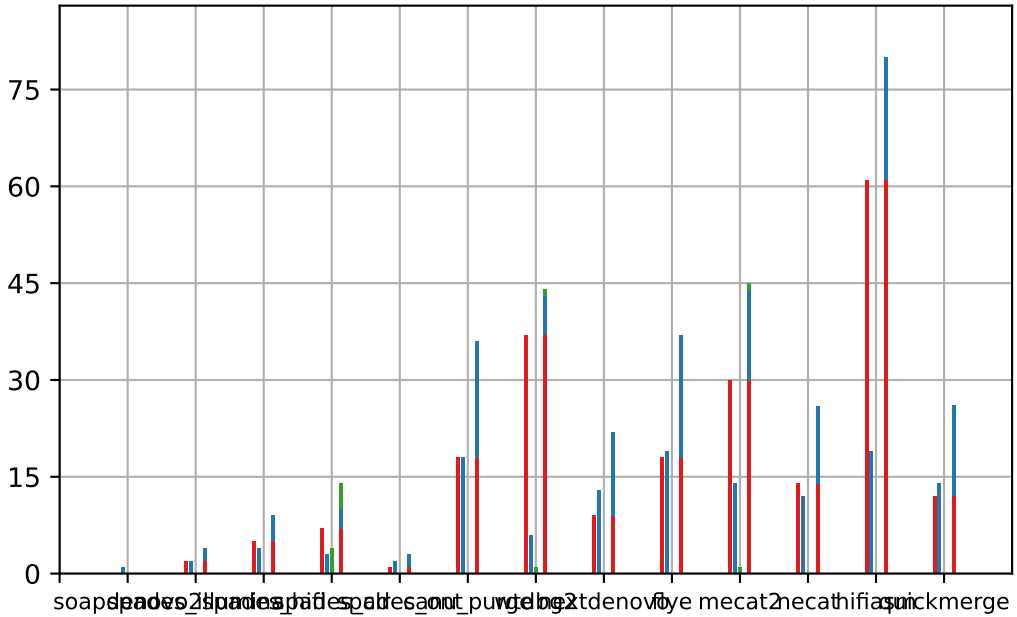


spades_ont coverage histogram (bin size: 1x)

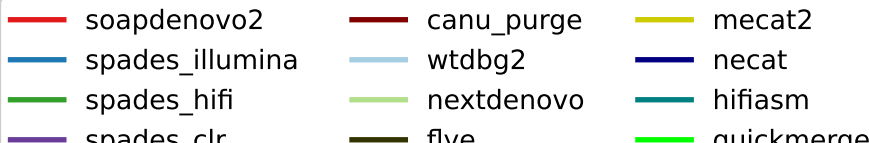
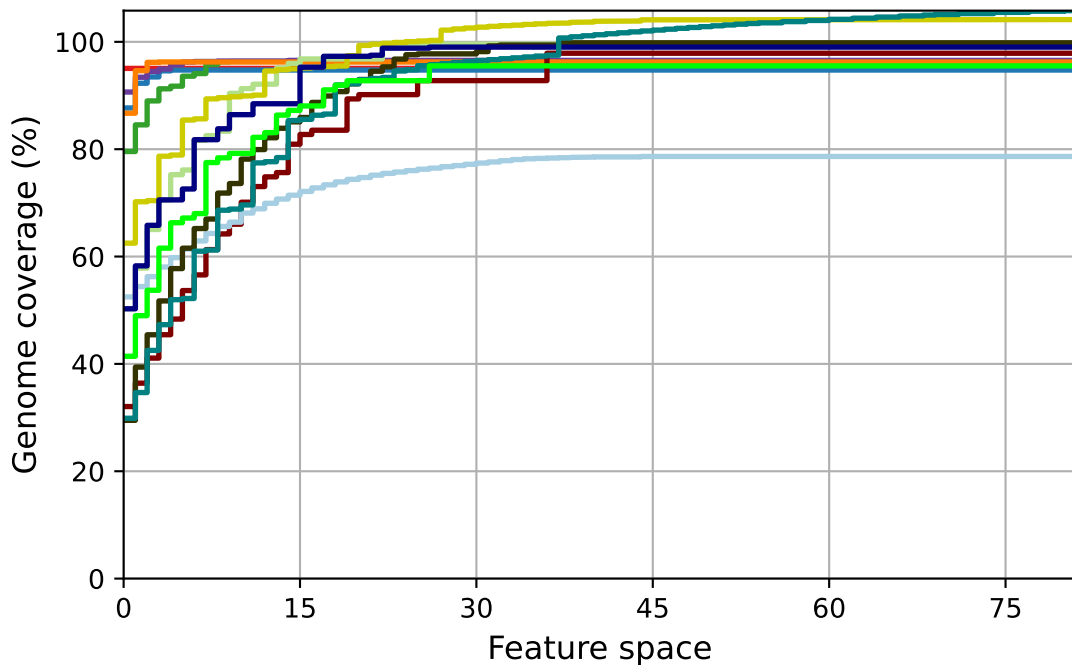


spades_ont

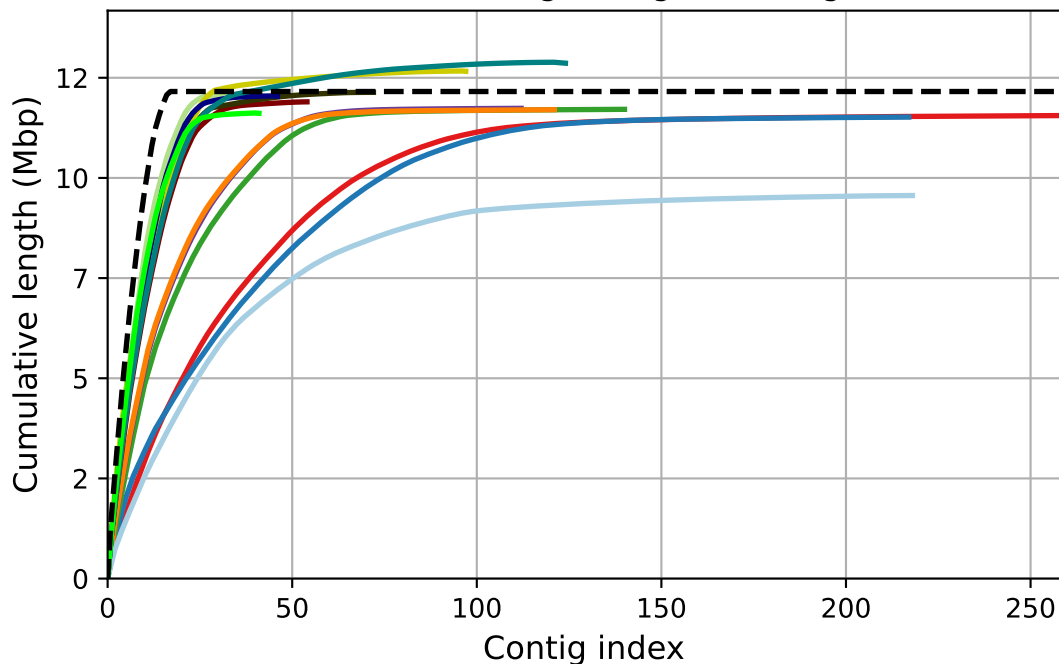
Misassemblies



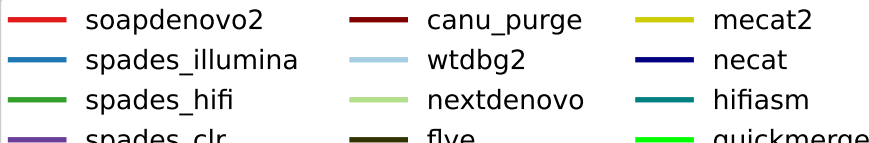
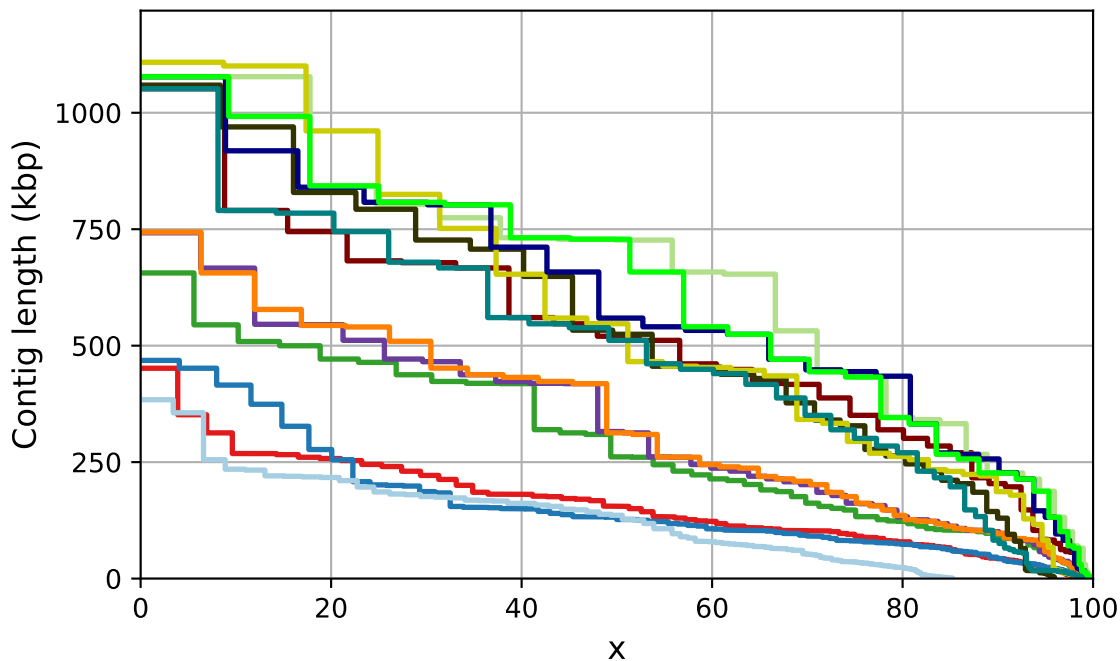
FRCurve (misassemblies)



Cumulative length (aligned contigs)



NAx



NGAx

